

Frequency Estimation in the Presence of Doppler Spread: Performance Analysis

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Abstract—In this paper, we are concerned with the estimation of the frequency of a complex sinusoid that has been corrupted by complex-valued multiplicative and additive noises. This problem is important in many applications including array processing in the case of spatially distributed sources and synchronization in the context of time-selective channels. The multiplicative noise smears the spectral line due to the sinusoid. This smearing, which is often called Doppler spreading, may significantly degrade the estimation accuracy. The goal of this paper is to analytically assess this degradation. The finite-sample Cramér–Rao bounds (CRBs) are derived, and closed-form expressions are given for the large-sample CRB. The latter gives insights into the *effective* coherent and non-coherent SNRs for frequency estimation. We then analyze the accuracy of frequency estimators that are based on the angles of the sample covariances. Simulations results are presented to illustrate the theoretical results.

Index Terms—Angular spread, Cramér–Rao lower bound, Doppler spread, fading, frequency estimation, multiplicative noise.

I. INTRODUCTION

FREQUENCY estimation in the presence of multiplicative noise is important for synchronization of communications signals in the context of time-selective fading channels. The fading-induced multiplicative noise is often referred to as short-scale fading [14, ch. 14]. In mobile communications, this fading is due to local scatterers around the mobile unit [15, Sec. 4.7]. The bandwidth of the fading process (the Doppler spread) increases with the speed of the mobile. Multiplicative noise is also encountered in coherent radar processing when the dynamics of the changes in the radar cross section are significant. The constant amplitude model is inadequate for pulse-to-pulse fluctuating targets [20]; these fluctuations contribute to a spectral spread of the received signal. This implies that the received pulses in a batch have neither the same amplitude nor the same initial phase. However, the phases of pulses are not uncorrelated, as in noncoherent radar processing. They are (highly) correlated, and the strength of this correlation is inversely proportional to the rate of the fluctuations. It is this correlation spread, which is sometimes identified as the scatterer “dwell” time, that makes

the phase information useful. Multiplicative noise models are also appropriate in array processing when the source signal is spatially distributed [19], as well as for backscattered acoustic signals [1] and SAR imagery systems.

A complex sinusoid that has been corrupted by fading and additive noise may be modeled as

$$x(n) = a(n)\exp(j\omega_o n) + v(n), \quad n = 0, 1, \dots, N-1 \quad (1)$$

where

- $\omega_o \in [-\pi, \pi)$ normalized frequency to be estimated;
- $a(n)$ fading-induced multiplicative noise;
- $v(n)$ models thermal noise and other interferences.

In the telecommunications context, model (1) is valid in the data-aided scenario when the symbol timing is perfectly recovered; ω_o represents the frequency offset due to mismatch between the received signal carrier and the local oscillator frequencies [5]. In Doppler radar, ω_o is the Doppler frequency that is related to the range/velocity of the target. In the array processing context, ω_o is the spatial frequency that is related to the direction of arrival (DOA) of the source, $a(n)$ models the spatial distribution of the source, and N is the number of sensors.

We make the following general assumptions:

- AS1) $a(n)$ is a complex-valued Gaussian process, circularly symmetric around its mean, $\mu e^{j\phi}$ ($\mu \geq 0$ and $-\pi \leq \phi < \pi$); its covariance function $R_a(\tau)$ is parameterized by a real-valued parameter vector $\alpha = [\alpha_1, \dots, \alpha_p]^T$. Let $\sigma_a^2 := R_a(0)$.
- AS2) Additive noise $v(n)$ is a zero-mean circular complex white Gaussian process with variance σ_v^2 .
- AS3) $a(n)$ and $v(n)$ are mutually independent.

Under AS1), the amplitude of the fading process will have a Rayleigh distribution if $\mu = 0$ and $\sigma_a^2 \neq 0$ and a Rician (dominant + Rayleigh) distribution if $\mu \neq 0$ and $\sigma_a^2 \neq 0$. The classical case of additive white Gaussian noise (AWGN) case is obtained when $\mu \neq 0$ and $\sigma_a^2 = 0$. It is common to define the Rician factor as the ratio of the power of the dominant path to that of the scattered path, i.e., $\kappa = \mu^2/\sigma_a^2$.

A common assumption for the fading amplitude $a(n)$ is the following.

- AS4) The covariance function $R_a(\tau)$ is real-valued and positive.

This assumption implies that the power spectral density of $a(n)$ is symmetric around zero frequency. Assumption AS4) has been made implicitly by many authors, e.g., in [14, ch. 14] for ionospheric channels and Doppler-radar and [15, Sec. 4.7] for mobile cellular communications. In the latter application, AS4) implies that the scatterers around the mobile unit are uniformly

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distributed in angle. The Doppler spread Δ_ω is a measure of the bandwidth (inverse coherence time) of the fading process. Using the Jakes model, the correlation of the fading process is given by $R_a(\tau) = \sigma_a^2 J_0(\Delta_\omega \tau)$, where $J_0(\cdot)$ is the zeroth-order Bessel function of the first kind [15, Sec. 4.7]. Other correlation models, such as Gaussian and exponential shapes, are used in ionospheric communications and in Doppler-radar [2]. Assumption AS4) could be relaxed if one could postulate a model for $R_a(\tau)$, whose parameters can be consistently identified from the observed data in (1) without knowledge of ω_o . In this case, the phase of $R_a(\tau)$ can be estimated; as we will see in Section III, this suffices to ensure consistent estimation of ω_o via the algorithm proposed in Section III.

It is interesting to note that the case of *real-valued* amplitude modulation [i.e., $a(n)$ is purely real] has been treated in great detail; see, e.g., [3] and [6]; the zero-mean complex Gaussian $a(n)$ is considered in [17] but under the restrictive assumption that the real and imaginary parts are iid, mutually uncorrelated, and have unequal variances. However, as noted in [1] and [7], the complex-valued amplitudes $a(n)$ should be modeled as circular Gaussian and should be temporally correlated. Indeed, the noncircular case offers an unrealistic extra degree of diversity and does not suffer from the *floor effect* (which is explained in Remark 3 following Proposition 1).¹

The focus of this paper is on the estimation of the frequency ω_o in model (1). We derive closed-form expressions for the Cramér–Rao bound (CRB). The effect of the Doppler spread on the CRB is investigated. The CRB expressions provide insights regarding the *effective* SNR for a sinusoidal signal in multiplicative noise. Recall that the CRB for the AWGN case has been studied in [16]. The CRB for the *real-valued* multiplicative noise model was derived in [6]. The real and complex-valued multiplicative noise models differ fundamentally in how the noises affect frequency estimation. Here, we address the *complex-valued* multiplicative noise scenario. We derive closed-form expressions for the finite and large-sample CRB. Expressions for the CRBs were also derived in [7]; many issues remained, however, to be addressed. For example, in this paper, we provide a proof that the estimation of the frequency and the noise parameters are decoupled (in [7], the multiplicative noise spectrum was assumed known). Furthermore, the approximate large sample CRB in [7] was developed for the strictly Rician case. Here, we give the *exact* CRB values, and our expressions are valid for both the Rayleigh and Rician cases. We will also show that even in the Rician case, the approximate CRB of [7] may not be very accurate. The CRB for the case of polynomial phase signals may be found in [5].

Frequency estimation in the presence of zero-mean complex-valued multiplicative noise was studied in [1]. Expressions for the variance of the so-called *pulse-pair* (PP) estimator were derived; this estimator involves estimating the phase of a single lag of the sample correlation. To assess the efficiency of the PP method, the author of [1] compared the error variance with the CRB for the *constant* amplitude case. Here, using our CRB formula, we show that the efficiency results established in [1] are

too pessimistic. In particular, we show that the PP method is *efficient* in the case of high signal-to-noise ratio (SNR), which is defined as

$$\text{SNR} := \frac{\mathcal{E}\{|a(n)|^2\}}{\mathcal{E}\{|v(n)|^2\}} = \frac{\mu^2 + \sigma_a^2}{\sigma_v^2}$$

where $\mathcal{E}$ denotes the mathematical expectation operator. In general, we show that the multiplicative noise deteriorates the CRB but does not significantly degrade the relative efficiency (with respect to the CRB) of the PP method. We also analyze algorithms that are based on averaging multiple sample covariance angles.

The structure of the paper is as follows. Section II is devoted to the computation of the finite and large-sample CRBs. The effect of the Doppler spread on the CRB is investigated and illustrated with some numerical examples. The frequency estimators under study are presented in Section III. The performance analysis of these estimators is provided in Section IV. Some simulation results are reported in Section V. Conclusions are presented in Sections VI, and Appendices follow.

II. CRAMÉR–RAO BOUNDS

In the Rician case, the unknown parameter vector for our model is

$$\boldsymbol{\theta} = [\omega_o, \phi, \mu, \boldsymbol{\alpha}^T, \sigma_v^2]^T.$$

In the Rayleigh case, $\mu = 0$, and the parameter vector reduces to

$$\boldsymbol{\theta} = [\omega_o, \boldsymbol{\alpha}^T, \sigma_v^2]^T.$$

The covariance matrix of any unbiased estimator $\hat{\boldsymbol{\theta}}$ cannot be lower than the CRB, and the inverse of the Fisher information matrix (FIM) [9]

$$\mathcal{E} \left\{ (\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}) (\hat{\boldsymbol{\theta}} - \boldsymbol{\theta})^T \right\} \geq \text{CRB}(\boldsymbol{\theta}) = \mathbf{J}_{\boldsymbol{\theta}, \boldsymbol{\theta}}^{-1} \quad (2)$$

where $A \geq B$ means that the matrix $A - B$ is positive semi-definite.

A. Finite-Sample CRB

Denote the fading amplitude and observed data vectors by

$$\mathbf{a} = [a(0), \dots, a(N-1)]^T; \quad \mathbf{x} = [x(0), \dots, x(N-1)]^T.$$

Vector $\mathbf{x}$ is Gaussian with mean

$$\mathbf{m}_x = \mu e^{j\phi} \Gamma \mathbf{1}$$

where $\mathbf{1}$ is a $(N \times 1)$ vector of ones, and

$$\Gamma = \text{diag} \{ e^{j\omega_o n}, n = 0, \dots, N-1 \}.$$

Let $\tilde{\mathbf{x}} = \mathbf{x} - \mathbf{m}_x$; since $\tilde{\mathbf{x}}$ is zero-mean and complex Gaussian, its statistics are completely characterized by the two correlation matrices

$$\mathbf{R}_x := \mathcal{E} \{ \tilde{\mathbf{x}} \tilde{\mathbf{x}}^H \}; \quad \mathbf{U}_x := \mathcal{E} \{ \tilde{\mathbf{x}} \tilde{\mathbf{x}}^T \}.$$

¹Noncircularity could result if the amplifiers associated with the I and Q paths are unbalanced, but this would lead to gross demodulation errors.

Since $a(n)$ and $v(n)$ are mutually independent, these matrices are given by

$$\mathbf{R}_x = \Gamma \mathbf{R}_a \Gamma^H + \sigma_v^2 \mathbf{I}; \quad \mathbf{U}_x = \Gamma \mathbf{U}_a \Gamma$$

where $\mathbf{R}_a = \mathcal{E}\{\tilde{\mathbf{a}}\tilde{\mathbf{a}}^H\}$, $\mathbf{U}_a = \mathcal{E}\{\tilde{\mathbf{a}}\tilde{\mathbf{a}}^T\}$, and $\mathbf{I}$ is the $(N \times N)$ identity matrix. Under assumption AS1), $a(n)$ is circular around its mean so that $\mathbf{U}_a$ vanishes [12], and the matrix $\mathbf{U}_x$ is then identically zero. This implies that $\tilde{\mathbf{x}}$ is *circularly symmetric*. The exact (finite-sample) FIM elements can then be written as [9, p. 525]

$$J_{\theta_k, \theta_\ell} = 2\mathcal{R} \left\{ \frac{\partial \mathbf{m}_x^H}{\partial \theta_k} \mathbf{R}_x^{-1} \frac{\partial \mathbf{m}_x}{\partial \theta_\ell} \right\} + \text{tr} \left\{ \mathbf{R}_x^{-1} \frac{\partial \mathbf{R}_x}{\partial \theta_k} \mathbf{R}_x^{-1} \frac{\partial \mathbf{R}_x}{\partial \theta_\ell} \right\} \quad (3)$$

where $\mathcal{R}(u)$ is the real part of u , and tr denotes the trace operator. The FIM entries are given in the following proposition.

Proposition 1: The finite-sample FIM for $\boldsymbol{\theta}$ under assumptions AS1)–AS3) is given by

$$\begin{aligned} J_{\omega_o, \omega_o} &= 2\mu^2 \mathbf{1}^T \mathbf{D} \mathbf{R}^{-1} \mathbf{D} \mathbf{1} + 2\text{tr} \{ \mathbf{R}^{-1} \mathbf{D} \mathbf{R} \mathbf{D} - \mathbf{D}^2 \} \\ J_{\omega_o, \phi} &= 2\mu^2 \mathcal{R} \{ \mathbf{1}^T \mathbf{R}^{-1} \mathbf{D} \mathbf{1} \} \\ J_{\omega_o, \alpha_k} &= -2\mathcal{I} \left\{ \text{tr} \left\{ \mathbf{D} \frac{\partial \mathbf{R}_a}{\partial \alpha_k} \mathbf{R}^{-1} \right\} \right\}, \quad k = 1, \dots, p \\ J_{\phi, \phi} &= 2\mu^2 \mathbf{1}^T \mathbf{R}^{-1} \mathbf{1} \\ J_{\mu, \mu} &= 2\mathbf{1}^T \mathbf{R}^{-1} \mathbf{1} \\ J_{\alpha_k, \alpha_\ell} &= \text{tr} \left\{ \mathbf{R}^{-1} \frac{\partial \mathbf{R}_a}{\partial \alpha_k} \mathbf{R}^{-1} \frac{\partial \mathbf{R}_a}{\partial \alpha_\ell} \right\}, \quad k, \ell = 1, \dots, p \\ J_{\alpha_k, \sigma_v^2} &= \text{tr} \left\{ \mathbf{R}^{-1} \frac{\partial \mathbf{R}_a}{\partial \alpha_k} \mathbf{R}^{-1} \right\}, \quad k = 1, \dots, p \\ J_{\sigma_v^2, \sigma_v^2} &= \text{tr} \{ \mathbf{R}^{-2} \} \end{aligned}$$

where $\mathbf{R} = \mathbf{R}_a + \sigma_v^2 \mathbf{I}$, $\mathbf{D} = \text{diag}\{0, 1, \dots, N-1\}$, and $\mathcal{I}(u)$ is the imaginary part of u . The other elements are zero. $\square$

Proof: see Appendix A.

Remarks:

- 1) Notice that the FIM does not depend on ω_o , even when the multiplicative noise is colored. This is in contrast with the case of colored additive noise [18]. In addition, note that the bound for μ , which is the amplitude of the dominant path, is decoupled from those for the rest.
- 2) Under AS4), we see from the third equation of the proposition that $J_{\omega_o, t > \alpha} = \mathbf{0}$. Hence, the FIM is block diagonal, and the CRBs for (ω_o, ϕ) , (μ) , and $(\boldsymbol{\alpha}^T, \sigma_v^2)$, i.e., the frequency and phase, the magnitude of the mean, and the covariance terms, are decoupled from one another. This observation provides justification for ignoring $\boldsymbol{\alpha}$ (which characterize the covariance matrix of the multiplicative noise) as in [7] (where the Doppler spectrum was assumed known to a scale factor). Thus, the CRBs are the same whether or not the covariance parameters of the multiplicative noise are known. Under AS4), the

CRBs in the Rayleigh and Rician cases are then, respectively, given by

$$\text{CRB}(\omega_o) = J_{\omega_o, \omega_o}^{-1}, \quad \text{CRB}(\omega_o, \phi) = J_{[\omega_o, \phi], [\omega_o, \phi]}^{-1}.$$

- 3) The expression for the FIM involves inversion of $N \times N$ matrices; the inversion can be done cheaply in the noiseless case ($\sigma_v^2/\sigma_a^2 \approx 0$) for the often used exponential correlation model, where $R_a(\tau) = \sigma_a^2 \exp(-\Delta_\omega |\tau|)$, and where Δ_ω is the Doppler spread. In this case, the first term of J_{ω_o, ω_o} , $J_{\omega_o, \phi}$, and $J_{\phi, \phi}$ can be computed via the approach given in [4]; the second term of J_{ω_o, ω_o} is found to be

$$2\text{tr} \{ \mathbf{R}_a^{-1} \mathbf{D} \mathbf{R}_a \mathbf{D} - \mathbf{D}^2 \} = \frac{2(N-1) \exp(-2\Delta_\omega)}{1 - \exp(-\Delta_\omega)}. \quad (4)$$

Thus, the potential accuracy of frequency estimation decreases with increasing Doppler spread. Apart from the AWGN case, the (finite-length) CRB for ω_o does not vanish when $\sigma_v^2 = 0$. This is known as the *floor effect*.

B. Asymptotic CRB

We next consider the asymptotic (i.e., large-sample) CRB under assumption AS4). The asymptotic CRB will be a good approximation of the exact CRB if the data record length N is *much larger than the coherence time* of $a(n)$ [i.e., effective duration of $R_a(k)$] (see [9, p. 535]). Under AS4), the CRBs for (ω_o, ϕ) and $(\boldsymbol{\alpha}^T, \sigma_v^2)$ are decoupled from one another; hence, we need only consider J_{ω_o, ω_o} , $J_{\omega_o, \phi}$, and $J_{\phi, \phi}$ for the derivation of the CRBs for ω_o and ϕ . In Appendix B, we derive the following Proposition.

Proposition 2: The large-sample FIM entries for the frequency, the phase, and mean parameters of the model in (1) operating under assumptions AS1)–AS4) are given by

$$\begin{aligned} J_{\omega_o, \omega_o} &\approx \left[\frac{2}{3} N^3 \beta + N \delta \right] \\ J_{\omega_o, \phi} &\approx N^2 \beta \\ J_{\phi, \phi} &\approx 2N \beta \\ J_{\mu, \mu} &= 2N \beta / \mu^2 \end{aligned}$$

where

$$\begin{aligned} \beta &:= \frac{\mu^2}{S_a(0) + \sigma_v^2} \\ \delta &:= \frac{1}{2\pi} \int_{-\pi}^{\pi} \left(\frac{S'_a(\omega)}{S_a(\omega) + \sigma_v^2} \right)^2 d\omega. \end{aligned}$$

Here, $S_a(\omega) := \sum_{\tau} R_a(\tau) \exp(-j\omega\tau)$ denotes the power spectral density (PSD) of the fading process, and $S'_a(\omega)$ is its derivative with respect to ω . The large-sample CRBs for ω_o , ϕ , and μ are given by

$$\begin{aligned} \text{CRB}(\omega_o) &\approx \frac{6}{N(N^2 \beta + 6\delta)} \\ \text{CRB}(\phi) &\approx \frac{1}{N\beta} \left[\frac{2N^2 \beta + 3\delta}{N^2 \beta + 6\delta} \right] \\ \text{CRB}(\mu) &\approx \frac{S_a(0) + \sigma_v^2}{2N}. \end{aligned}$$

$\square$

Remarks:

- 1) Parameter β can be interpreted as the *effective* coherent signal-to-noise ratio (SNR) and δ as the *effective* noncoherent SNR, which is a measure of deviation from flatness of the spectrum. Parameter δ completely captures the large-sample effect of the color of the multiplicative noise.
- 2) If $S'_a(\pm\pi) = 0$, which is often the case since $a(n)$ is a narrowband lowpass process, δ can be rewritten as

$$\delta = \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{S''_a(\omega)}{S_a(\omega) + \sigma_v^2} d\omega$$

where $S''_a(\omega)$ is the second derivative of $S_a(\omega)$ with respect to ω . This expression shows how performance depends on the second derivative of the spectrum of the multiplicative noise.

- 3) In the AWGN case, $S_a(\omega) \equiv 0$, $\delta = 0$ and $\beta = \mu^2/\sigma_v^2$, and the FIM reduces to that derived in [9, p. 542] and [13, p. 292].
- 4) In the case of flat fading, $S'_a(\omega) = 0$, so that $\delta = 0$, and $\beta = \mu^2/(\sigma_a^2 + \sigma_v^2)$, and the CRB for ω_o is proportional to $1/\beta$:

$$\text{CRB}(\omega_o) \approx \frac{6}{N^3\beta} = \frac{6}{N^3} \frac{\sigma_a^2 + \sigma_v^2}{\mu^2}.$$

It is remarkable that the CRB for the case of flat fading and $\mu \neq 0$ is a scalar multiple of that in the constant amplitude case. Note that the CRB has the N^{-3} form in both cases.

- 5) The term in square brackets in the expression for $\text{CRB}(\phi)$ is of order $O(1)$ since it is lower and upper bounded by $1/2$ and 2 , respectively.
- 6) For Rayleigh fading, $\mu = 0$; hence, $\beta = 0$, and since $\mu = 0$, we may take $\phi = 0$ without loss of generality (wlog). Hence

$$\text{CRB}(\omega_o) \approx \frac{1}{N\delta}.$$

Now, the CRB is $O(N^{-1})$, i.e., two orders of magnitude more than those in the constant amplitude and nonzero mean cases. This is in contrast with the *real-valued* multiplicative noise (or amplitude noise) model for which the CRB decreases at the rate $O(N^{-3})$ [6]. This is because phase noise (which is introduced by the complex-valued multiplicative noise) has more deleterious effects on frequency estimation than does amplitude noise. To illustrate the difference, let us consider the purely multiplicative noise for which the additive noise $v(n) = 0$. With $\mu = 0$, we have $x_c(n) = |a(n)|e^{j\omega n + j\angle\{a(n)\}}$ if $a(n)$ is complex-valued, and $x_r(n) = |a(n)|e^{j\omega n + jk(n)\pi}$, with $k = 0$ or 1 , if $a(n)$ is real valued. Notice that ω_o can be estimated *perfectly* from $x_r(n)$ because of the diversity afforded by the noncircularity of $a(n)$.

- 7) If the fading process is both zero-mean and flat, then $\delta = 0$ and $\beta = 0$ so that the FIM $\mathbf{J} = 0$; hence, none of the parameters is identifiable.

- 8) We stress that Proposition 2 gives a closed-form expression for the asymptotic FIM and asymptotic CRB. In [7], the FIM entries were left as an infinite summation.
- 9) An approximation of the asymptotic CRB (ACRB) was developed in [7] for the strictly Rician case ($\mu \neq 0$). We give the *exact* ACRB values, and our expressions are valid even when $\mu = 0$. Even in the nonzero mean case, the approximate ACRB of [7] may not be very accurate; for narrowband process, the δ term in our exact ACRB may be very large so that the $N^2\beta$ term may not be (significantly) larger than δ . Further, we capture the influence of the shape of the spectrum of the fading process on the performance bounds via a single parameter δ .
- 10) Since the FIM for (ω_o, ϕ) , (μ) and (α^T, σ_v^2) are decoupled, the CRB for the latter is given by

$$\text{CRB}(\alpha, \sigma_v^2) = J_{[\alpha, \sigma_v^2]}^{-1}[\alpha, \sigma_v^2]$$

where the FIM entries can be obtained using Whittle's formula (see Appendix B).

- 11) If $R_a(\tau)$ is not real-valued, i.e., $S_a(\omega)$ is not symmetric around $\omega = 0$, the CRB's for ω_o and α are not decoupled, *not even asymptotically*. Indeed, the large-sample expression

$$J_{\omega_o, \alpha_k} \approx \frac{N}{2\pi} \int_{-\pi}^{\pi} \frac{-S'_a(\omega) \frac{\partial S_a(\omega)}{\partial \alpha_k}}{(S_a(\omega) + \sigma_v^2)^2} d\omega$$

is zero only if $S_a(\omega)$ is symmetric around $\omega = 0$ [which implies that $S'_a(\omega)$ is antisymmetric]. Hence, if AS4) is not satisfied, one should compute all the FIM entries in order to derive the CRB for ω_o . This implies that the maximum-likelihood estimation of ω_o and the parameters of the fading process are, in general, not decoupled.

- 12) The FIM in the case of *colored* additive noise can be computed along the lines of the uncorrelated noise case. The FIM is, however, not block diagonal even under assumption AS4). One thus needs to compute all entries of the FIM and, then, its inverse in order to evaluate the CRB for ω_o .

C. Effect of the Doppler Spread on the CRB

The Doppler spread Δ_ω is the effective bandwidth of the random process, $S_a(\omega)$; from the definition of δ , we see that δ depends on both $S_a(\omega)$ as well as its derivative; hence, the dependence of $\text{CRB}(\omega_o)$ on Δ_ω is very nonlinear, particularly in the Rician case $\mu \neq 0$. In the case of exponentially decaying correlation and Rayleigh fading ($\kappa = 0$), we see from (4) that $\text{CRB}(\omega_o)$ increases with the Doppler spread Δ_ω . This monotonicity does not hold in the Rician case, particularly for large κ . To illustrate the dependence of the CRB on Δ_ω , we study the Gaussian-shaped $S_a(\omega)$,

$$\begin{aligned} S_a(\omega) &= \frac{2\sigma_a^2\sqrt{\pi}}{\Delta_\omega} \exp[-(\omega/\Delta_\omega)^2] \longleftrightarrow R_a(\tau) \\ &= \sigma_a^2 e^{-\Delta_\omega^2 \tau^2/4} \end{aligned} \quad (5)$$

which is a function of two parameters: the variance σ_a^2 and the Doppler spread Δ_ω . This is sometimes used as an approxima-

tion to the Clarke model [15, Sec. 4.7] when the normalized Doppler spread is small. Other spectral models, such as the exponential shape, can be studied similarly. The large-sample CRB for the Jakes model cannot be derived since the corresponding spectrum is not differentiable, and thus, δ is not defined.

In the noiseless case, i.e., $\sigma_v^2 \approx 0$, we have

$$\delta|_{\sigma_v^2=0} = \frac{4\pi^2}{3(\Delta_\omega)^4}.$$

Hence, as expected, the noncoherent SNR δ decreases as the Doppler spread Δ_ω increases. The coherent SNR β is given by

$$\beta|_{\sigma_v^2=0} = \frac{\mu^2}{S_a(0)} = \frac{\mu^2}{2\sqrt{\pi}\sigma_a^2} \Delta_\omega = \kappa \frac{\Delta_\omega}{2\sqrt{\pi}}$$

which, unlike δ , increases with Δ_ω . This is in agreement with the large-sample CRB for the frequency of a constant amplitude harmonic in colored noise, which is affected only by the local SNR (see [4], and references therein); in our case, the local SNR is equal to $\mu^2/S_a(0)$, and $S_a(0)$ decreases as Δ_ω increases.

The CRB for the frequency is therefore given by

$$\text{CRB}(\omega_o)|_{\sigma_v^2=0} \approx \frac{6}{\frac{\kappa \Delta_\omega}{2\sqrt{\pi}} N^3 + \frac{8\pi^2}{\Delta_\omega^4} N}.$$

This CRB first increases until $\Delta_\omega = 2\sqrt{\pi}(2/\kappa N^2)^{1/5}$, beyond which, the CRB starts decreasing slightly. Thus, in the Rician case, the CRB does not necessarily increase with increasing Doppler spread. This result is somewhat surprising since the Doppler spread has always been considered to be a nuisance parameter.²

With $\alpha = [\Delta_\omega]$, from Proposition 1, and Whittle's formula [13, Th. 5.3], we have

$$\begin{aligned} J_{\Delta_\omega, \Delta_\omega} &= \text{tr} \left\{ \mathbf{R}^{-1} \frac{\partial \mathbf{R}_a}{\partial \Delta_\omega} \mathbf{R}^{-1} \frac{\partial \mathbf{R}_a}{\partial \Delta_\omega} \right\} \\ &\longrightarrow N \frac{1}{2\pi} \int_{-\pi}^{\pi} \left(\frac{\partial \mathbf{S}_a(\omega)}{\partial \Delta_\omega} \right)^2 d\omega. \end{aligned}$$

Assuming that $\Delta_\omega/(2\pi) \ll 1$, the large-sample CRB for Δ_ω is given by

$$\begin{aligned} \text{CRB}(\Delta_\omega)|_{\sigma_v^2=0} &\approx \frac{\Delta_\omega^2}{N} \left[\frac{4}{5} \frac{\pi^4}{\Delta_\omega^4} - \frac{4}{3} \frac{\pi^2}{\Delta_\omega^2} + 1 \right]^{-1} \\ &\approx \frac{1}{N} \frac{5}{4\pi^4} \Delta_\omega^6 \end{aligned}$$

where the last approximation holds for small Δ_ω ; note that the CRB increases with Δ_ω . In the noisy case, the behavior of the CRB is similar to that in the noiseless case; however, δ can be computed only via numerical integration.

²The threshold Δ_ω varies as $N^{-2/5}$; hence, for large N , the threshold is close to zero, and the CRB decreases as the Doppler spread increases.

III. FREQUENCY ESTIMATION

If the fading process $a(n)$ has nonzero mean, or if its spectrum is unimodal with known mode [typically, $a(n)$ is lowpass and its spectrum peaks are at DC], then the frequency ω_o can be estimated by locating the peak of the periodogram. This estimator is the maximum likelihood (ML) estimator in the constant amplitude case. In practice, this frequency domain technique is too time consuming, and simpler methods are preferable. The ML estimator for the multiplicative noise model in (1) is even more complex since it requires a multidimensional optimization procedure. The Pulse-Pair (PP) method is a good alternative because its computational load is low. The PP method was studied in [1] for the multiplicative noise model in (1) with $\kappa = 0$, and an expression for the asymptotic variance of the estimator was derived. Quoting from [1], “due to the multiplicative noise, the performance of the PP technique deteriorates and the estimation variance is much larger than the CRB.” This statement is too pessimistic. Indeed, the author of [1] compared the estimation variance with the CRB that corresponds to the constant amplitude case ($\kappa = \infty$). According to Section II-C, the CRB for the $\kappa = 0$ case might be much higher than that for the $\kappa \neq 0$ case (see Figs. 1–4 in Section V).

The PP method is a simple and computationally attractive method for the estimation of the frequency ω_o . The theoretical autocorrelation sequence of the noisy data in (1) is given by

$$\begin{aligned} c_x(m) &:= E\{x^*(n)x(n+m)\} \\ &= [R_a(m) + \mu^2] \exp(j\omega_o m) + \sigma_v^2 \delta_m. \end{aligned}$$

Thus, with $m > 0$, the frequency ω can be estimated from $c_x(m)$, provided that $r_a(m)$ satisfies assumption AS4), and $|\omega_o| \leq \pi/m$. To handle the general case of $|\omega_o| < \pi$, we must unwrap the phase. Based on this, the PP estimator is obtained as, with $m > 0$

$$\hat{\omega}_o^{(m)} = \frac{1}{m} \angle [\hat{c}_x(m)] \quad (6)$$

where $\hat{c}_x(m)$ is the sample autocorrelation

$$\hat{c}_x(m) = \frac{1}{N-m} \sum_{n=0}^{N-1-m} x^*(n)x(n+m) \quad (7)$$

and $\angle$ denotes the phase angle operator. Notice that $\hat{\omega}_o^{(m)}$ is unbiased under AS4).

In the nonfading case (constant amplitude sinusoid), it was shown that the optimal value of m minimizing the variance of the estimate in (6) is $2N/3$ [10]. We will see that this result does not hold true in the fading case. Indeed, the multiplicative noise decorrelates the signal so that the estimate $\angle\{\hat{c}_x(m)\}$ tends to become worse as m increases. This implies that m_o , which is the optimal value of m , should be lower than $2N/3$. According to the above observations, m_o should decrease with the Doppler spread of the fading process.

IV. PERFORMANCE ANALYSIS

In this section, we study the effect of the Doppler spread on the performance of the frequency estimators in (6) and (20). Toward this objective, we first derive expressions for the covari-

ances of the sample correlation and those of their phase angles. The following performance analysis will be useful for any frequency estimation method that is based on the sample correlations.

A. Sample Correlation Estimates

Proposition 3: Under assumptions AS1)–AS4), the finite-sample covariances of the sample autocorrelation estimates $\hat{c}_x(m)$ and $\hat{c}_x(\ell)$, $\ell \geq m > 0$ are given by

$$\begin{aligned} \text{cov}\{\hat{c}_x(m), \hat{c}_x(\ell)\} &= \frac{e^{j\omega_o(m-\ell)}}{(N-m)(N-\ell)} \\ &\cdot \left\{ \left(\sum_{\tau=m-N+1}^{N-1-\ell} D(\tau) R_a(\tau) R_a(\tau+\ell-m) \right. \right. \\ &\quad \left. \left. + D(\tau) [\mu^2 R_a(\tau) + \mu^2 R_a(\tau+\ell-m)] \right) \right. \\ &\quad \left. + 2D(0) \sigma_v^2 [\mu^2 + R_a(\ell-m)] + \sigma_v^4 D(0) \delta(\ell-m) \right\} \end{aligned} \quad (8)$$

$$\begin{aligned} \text{cov}\{\hat{c}_x(m), \hat{c}_x^*(\ell)\} &= \frac{e^{j\omega_o(m+\ell)}}{(N-m)(N-\ell)} \\ &\cdot \left\{ \left(\sum_{\tau=m-N+1}^{N-1-\ell} D(\tau) R_a(\tau-m) R_a(\tau+\ell) \right. \right. \\ &\quad \left. \left. + D(\tau) [\mu^2 R_a(\tau-m) + \mu^2 R_a(\tau+\ell)] \right) \right. \\ &\quad \left. + 2D(m) \sigma_v^2 [\mu^2 + R_a(\ell+m)] \right\} \end{aligned} \quad (9)$$

where

$$D(\tau) = \max(N-\ell-|\tau| - \min(0, \max(\tau, m-\ell)), 0). \quad (10)$$

Proof: See Appendix C

Note that $D(\tau)$ is actually a function of N , ℓ , and m ; for ease of notation, we do not explicitly show the dependence on these parameters.

We next derive the large-sample expressions for these covariances. By large-sample, we mean that $N \gg 1$ and $m, \ell \ll N$. We also assume that $R_a(\cdot)$ is absolutely summable

$$\sum_{n=-\infty}^{\infty} |R_a(\tau)| < \infty \quad (11)$$

which implies the continuity of $S_a(f)$. Let

$$\bar{R}_a(\tau) := \frac{1}{2\pi} \int S_a^2(\omega) e^{j\omega\tau} d\omega.$$

The computation of $\bar{R}_a(m)$ can be handled easily for many spectral models. For example, in the case of the Gaussian shaped

spectrum, we obtain the following closed-form expression:

$$\bar{R}_a(\tau) = \frac{\sigma_a^4 \sqrt{2\pi}}{\Delta_\omega} e^{-\Delta_\omega^2 \tau^2 / 8}.$$

Furthermore, when $a(n)$ is modeled as a linear process, $\bar{R}_a(m)$ can be computed easily since $S_a^2(\omega)$ is the spectrum of a linear process as well.

Under (11), we have

$$\frac{1}{N} \sum_{\tau} D(\tau) R_a(\tau) R_a(\tau+k) \xrightarrow{N \rightarrow \infty} S_a(0) \quad (12)$$

$$\frac{1}{N} \sum_{\tau} D(\tau) R_a(\tau) R_a(\tau+k) \xrightarrow{N \rightarrow \infty} \bar{R}_a(k). \quad (13)$$

The following proposition is therefore readily obtained.

Proposition 4: Under assumptions AS1)–AS4) and (11), the large-sample covariances of the sample autocorrelations $\hat{c}_x(m)$ and $\hat{c}_x(\ell)$, with $N \gg \ell \geq m > 0$, are given by

$$\begin{aligned} N \text{cov}\{\hat{c}_x(m), \hat{c}_x(\ell)\} &= e^{j\omega_o(m-\ell)} [\bar{R}_a(\ell-m) + 2\mu^2 S_a(0) + 2\sigma_v^2 \\ &\quad \cdot [\mu^2 + R_a(\ell-m)] + \sigma_v^4 \delta(\ell-m)] \end{aligned} \quad (14)$$

$$\begin{aligned} N \text{cov}\{\hat{c}_x(m), \hat{c}_x^*(\ell)\} &= e^{j\omega_o(m+\ell)} [\bar{R}_a(\ell+m) + 2\mu^2 S_a(0) + 2\sigma_v^2 \\ &\quad \cdot [\mu^2 + R_a(\ell+m)]] \end{aligned} \quad (15)$$

B. Phase Estimation

Let $\hat{\varphi}_m = \angle[\hat{c}_x(m)]$ denote the phase angle of $\hat{c}_x(m)$ in $[-\pi, \pi)$. Since $\angle$ is a nonlinear operator, the finite-sample error analysis for the phase estimation is, in general, not tractable. Here, we assume that the phase estimate is accurate enough and focus on the small-error analysis using the first-order truncation of the Taylor expansion.

Using Proposition 3, we derive the following proposition.

Proposition 5: Under assumptions AS1)–AS4), the finite-sample covariance of $\hat{\varphi}(m)$ and $\hat{\varphi}(\ell)$, $\ell \geq m > 0$ is given by

$$\begin{aligned} \text{cov}\{\hat{\varphi}(m), \hat{\varphi}(\ell)\} &= \frac{1}{2(N-\ell)(N-m)} \frac{1}{[\kappa + \rho_a(m)][\kappa + \rho_a(\ell)]} \\ &\cdot \left\{ \left(\sum_{\tau=m-N+1}^{N-1-\ell} D(\tau) [\rho_a(\tau) \rho_a(\tau+\ell-m) \right. \right. \\ &\quad \left. \left. - \rho_a(\tau+\ell) \rho_a(\tau-m)] + D(\tau) \right. \right. \\ &\quad \left. \left. \cdot \kappa [\rho_a(\tau) + \rho_a(\tau+\ell-m) - \rho_a(\tau+\ell) - \rho_a(\tau-m)] \right) \right. \\ &\quad \left. + 2D(0) \left(\frac{\kappa+1}{\text{SNR}} \right) [\kappa + \rho_a(\ell-m)] \right. \\ &\quad \left. - 2D(m) \left(\frac{\kappa+1}{\text{SNR}} \right) [\kappa + \rho_a(\ell+m)] \right. \\ &\quad \left. + \left(\frac{\kappa+1}{\text{SNR}} \right)^2 D(0) \delta(\ell-m) \right\} \end{aligned} \quad (16)$$

where the Rice factor, SNR, and normalized correlation are defined as

$$\kappa := \frac{\mu^2}{\sigma_a^2}, \quad \text{SNR} := \frac{\mu^2 + \sigma_a^2}{\sigma_v^2} = \frac{\kappa + 1}{\sigma_v^2/\sigma_a^2};$$

$$\rho_a(\tau) := \frac{R_a(\tau)}{\sigma_a^2}.$$

Proof: see Appendix D.

In Appendix E, we provide another approximation which is valid for smaller N . Using (12) and (13), we derive the following large-sample error expressions.

Proposition 6: Under assumptions AS1)–AS4) and (11), the large-sample covariance of $\hat{\varphi}(m)$ and $\hat{\varphi}(\ell)$, $N \gg \ell \geq m > 0$ is given by

$$\begin{aligned} \text{cov}\{\hat{\varphi}(m), \hat{\varphi}(\ell)\} &= \frac{1}{2N} \frac{1}{[\kappa + \rho_a(m)][\kappa + \rho_a(\ell)]} \\ &\cdot \left\{ [\bar{\rho}_a(\ell - m) - \bar{\rho}_a(\ell + m)] \right. \\ &\quad + 2 \left(\frac{\kappa + 1}{\text{SNR}} \right) [\rho_a(\ell - m) - \rho_a(\ell + m)] \\ &\quad \left. + \left(\frac{\kappa + 1}{\text{SNR}} \right)^2 \delta(\ell - m) \right\} \end{aligned} \quad (17)$$

where $\bar{\rho}_a(\cdot) = \bar{R}_a(\cdot)/\sigma_a^4$.

Notice that the large-sample variance decreases as $1/N$ regardless of the value of κ .

C. Performance of Frequency Estimators

With $m = 1$, the frequency can be unambiguously estimated; for higher values of m , the phase must be unwrapped. Here, we assume that the unwrapping is done properly, which is the case for large SNR/number of samples. We focus on the small error analysis of the frequency estimators. The variance expression of the PP estimator is then

$$\text{var}\{\hat{\omega}_o^{(m)}\} = \frac{1}{m^2} \text{var}\{\hat{\varphi}(m)\} \quad (18)$$

where $\text{var}\{\hat{\varphi}(m)\}$ is given in Proposition 5. The following cases are of interest.

- In the AWGN case, i.e., $\kappa = \infty$, the finite-sample variance of the PP estimator reduces to

$$\begin{aligned} \text{var}\{\hat{\omega}_o^{(m)}\} &= \frac{\min(m, N - m)}{m^2(N - m)^2} \frac{1}{\text{SNR}} \\ &\quad + \frac{1}{2m^2(N - m)} \frac{1}{\text{SNR}^2} \end{aligned}$$

which coincides with the variance expression given in [10]; in addition, recall that the optimal m is $m_o = 2N/3$ and independent of SNR.

- In the Rayleigh case, i.e., $\kappa = 0$, the large-sample variance turns out to be

$$\begin{aligned} \text{var}\{\hat{\omega}_o^{(m)}\} &= \frac{1}{2Nm^2\rho_a^2(m)} \\ &\cdot \left[[\bar{\rho}_a(0) - \bar{\rho}_a(2m)] + \frac{2}{\text{SNR}} [1 - \rho_a(2m)] \right. \\ &\quad \left. + \frac{1}{\text{SNR}^2} \right]. \end{aligned} \quad (19)$$

In contrast with the AWGN case, the rate of convergence is only $1/N$ for fixed m . For fixed m , the rate is not $1/m^2$ since the rate also depends on $\rho_a(m)$ and $\bar{\rho}_a(m)$; in typical models, these decay exponentially with m . Consequently, the variance expression typically attains a minimum at a small value of m , i.e., the optimal m_o is small.

The accuracy of the PP estimator may be improved by averaging estimates obtained with different values of m :

$$\hat{\omega}_o = \sum_{m=1}^M h(m) \angle [\hat{c}_x(m)], \quad M < N. \quad (20)$$

The variance of the estimator in (20) is obtained as

$$\text{var}\{\hat{\omega}_o\} = \mathbf{h}^T \mathbf{\Psi} \mathbf{h}$$

where $\mathbf{h} = [h(1), \dots, h(M)]^T$, and $\mathbf{\Psi}$ is the $M \times M$ covariance matrix of the estimated phase angle vector $\hat{\boldsymbol{\varphi}} = [\hat{\varphi}(1), \dots, \hat{\varphi}(M)]^T$, whose (k, ℓ) th elements are given in Proposition 5. The optimal weighting vector $\mathbf{h}_o$ for (20) is given by

$$\mathbf{h}_o = (\mathbf{u}^T \mathbf{\Psi}^{-1} \mathbf{u})^{-1} \mathbf{\Psi}^{-1} \mathbf{u}$$

where $\mathbf{u} = [1, \dots, M]^T$. The variance of this optimal estimator is

$$\text{var}\{\mathbf{h}_o^T \hat{\boldsymbol{\varphi}}\} = (\mathbf{u}^T \mathbf{\Psi}^{-1} \mathbf{u})^{-1}.$$

The practical relevance of such an estimator is minor because it requires the multiplicative and additive noise parameters to be known. However, its variance expression represents a lower bound on the variance of any unbiased phase-based frequency estimator of the form in (20).

V. SIMULATION RESULTS

We first study the accuracy of the PP estimator in (6) as a function of m . The fading process is assumed to have covariance function, $R_a(\tau) = \sigma_a^2 \exp(-\Delta_\omega |\tau|)$; the number of samples is fixed to $N = 20$. Fig. 1(a) displays m_o/N , m_o being the optimal value of m (for which the variance is minimum) versus the normalized Doppler bandwidth $N\Delta_\omega/(2\pi)$ for different values of the SNR in the Rayleigh case $\kappa = 0$ for which the variance expression is given in (19).

As expected, m_o/N decreases as $N\Delta_\omega$ increases. For low values of $N\Delta_\omega$, $m_o/N = 2/3$, as in the AWGN case. The ratio m_o/N starts decreasing when $N\Delta_\omega = 0.02$, and m_o eventually

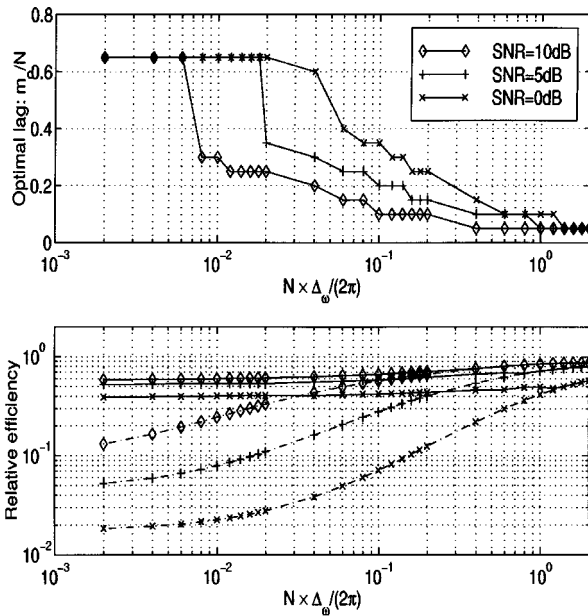


Fig. 1. Top: Optimal value m_o of m versus Doppler spread. Bottom: Relative efficiencies of $\hat{\omega}^{(1)}$ (dashed line) and of $\hat{\omega}^{(m_o)}$ (solid line) versus Doppler spread. Curves parameterized by different SNRs for an exponential shaped fading model; $\kappa = 0$.

approaches 1. If the SNR and Δ_ω are (approximately) known, m_o can be derived using (18).

Fig. 1(b) displays the corresponding relative efficiencies (RE) of $\hat{\omega}^{(m_o)}$ and $\hat{\omega}^{(1)}$ with regard to the CRB. It is seen that these REs are comparable at high SNR, suggesting that the simple estimator $\hat{\omega}^{(1)}$ could be used. However, for moderate to low SNRs, the RE of $\hat{\omega}^{(m_o)}$ is significantly higher than that of $\hat{\omega}^{(1)}$. The RE of both estimators increases with increasing Doppler bandwidth. The RE of $\hat{\omega}^{(m_o)}$ is not far from 1 for all values of Δ_ω , i.e., independent of the Doppler spread, the PP method has acceptable performance compared with the CRB. We remark that the ML estimator for this problem is highly nonlinear; good initial estimates can be provided via the low-complexity PP estimator and that the MLE is only asymptotically efficient. The same behavior is noticed when the Gaussian shaped correlation function is used, i.e., $R_a(\tau) = \sigma_a^2 \exp(-\Delta_\omega^2 \tau^2 / 4)$.

We next check the validity of the theoretical variance expressions of the estimator in (18) via Monte-Carlo simulations. The multiplicative noise is generated according to the exponential correlation model; the frequency was $\omega_o = 0.01\pi$. For different values of SNR, κ , and Δ_ω , we evaluate the empirical mean square errors (MSE), the theoretical variance, and the corresponding asymptotic CRBs. The empirical MSEs are obtained using 500 Monte Carlo runs.

Figs. 2 and 3 show the performance of the estimator for the case of slow fading ($\Delta_\omega = (2\pi)0.003$) and fast fading ($\Delta_\omega = (2\pi)0.01$) for different values of SNR, with $\kappa = 0$ (Rayleigh fading) for an exponential correlation model and $N = 1000$. The straight lines depict the CRBs, the os , which are the theoretical variances, and the x s, which are the empirical variances; solid lines correspond to higher SNRs and dashed lines to lower SNRs. Notice that at lower SNRs, the theoretical variance curve has a minimum around $m = 10$ for slow fading, and $m = 4$ for faster fading; for higher SNRs, the optimal values are $m = 2$

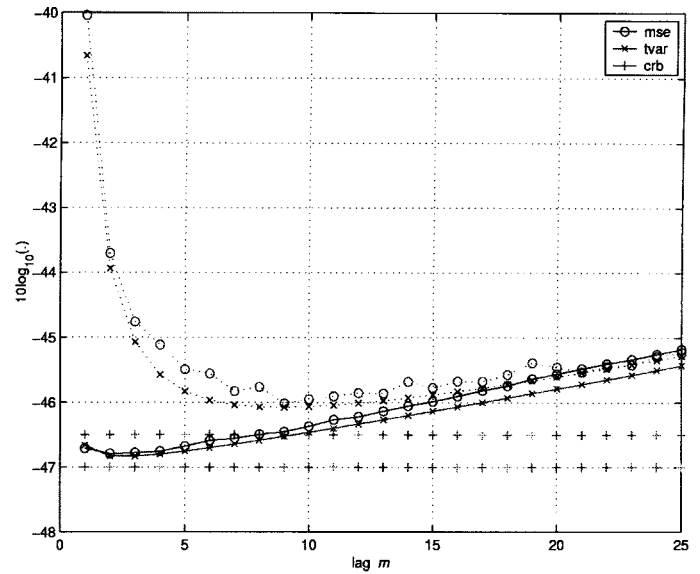


Fig. 2. Empirical MSE, theoretical variance, and CRB of $\omega^{(m)}$ on $\log_{10}$ scale versus m ; $\Delta_\omega = (2\pi)0.003$ (slow fading); $\kappa = 0$; $N = 1000$; SNR = 15 dB in solid lines, SNR = 5 dB in dotted lines.

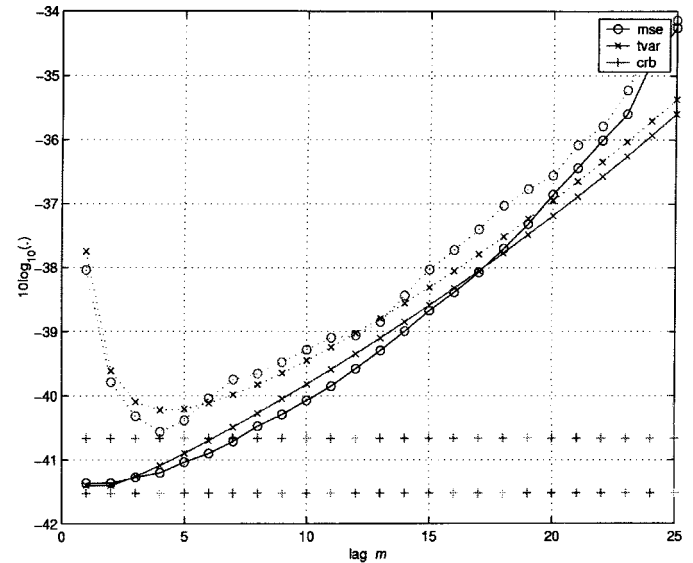


Fig. 3. Empirical MSE, theoretical variance, and CRB of $\omega^{(m)}$ on $\log_{10}$ scale versus m ; $\Delta_\omega = (2\pi)0.01$ (fast fading); $\kappa = 0$; $N = 1000$; SNR = 20 dB in solid lines, SNR = 5 dB in dotted lines.

and $m = 1$. The empirical curves are close to the theoretical ones; however, the estimators become inefficient as m differs from m_o , i.e., larger m .

For $m > m_o$, the variance of the estimator increases with m . To understand this, it is instructive to look at the special case of Rayleigh fading: The term $m^2 \rho_a^2(m)$ in the denominator explains why the variance increases with m ; $\rho_a(m)$, which is the temporal correlation, decays (exponentially in the model used in the simulations) offsetting the gain due to the m term.

In Figs. 4–8, we plot the empirical performance, the theoretical performance, and the asymptotic CRB versus SNR, the Rician parameter κ , the number of samples N , the Doppler spread Δ_ω , and the frequency ω_o respectively. The relevant parameters were exponentially shaped correlation with $\Delta_\omega = (2\pi)0.01$,

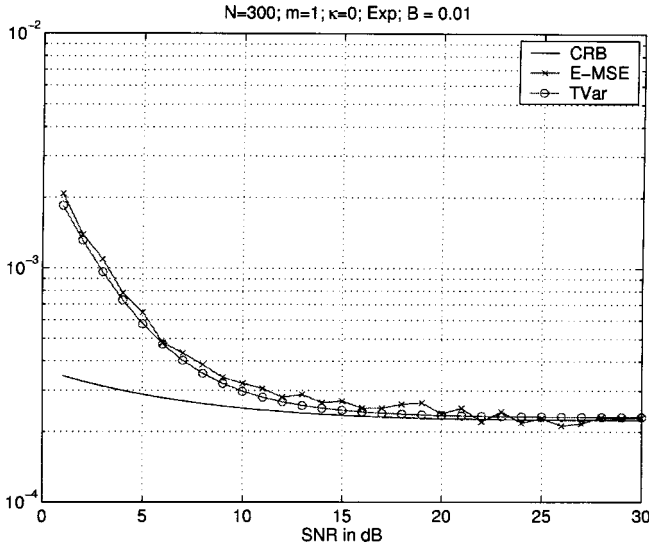


Fig. 4. Empirical MSE, theoretical variance, and CRB of $\omega^{(m)}$ versus SNR; $m = 1$; $\Delta_\omega = (2\pi)0.01$ (fast fading); $\kappa = 0$; $N = 300$.

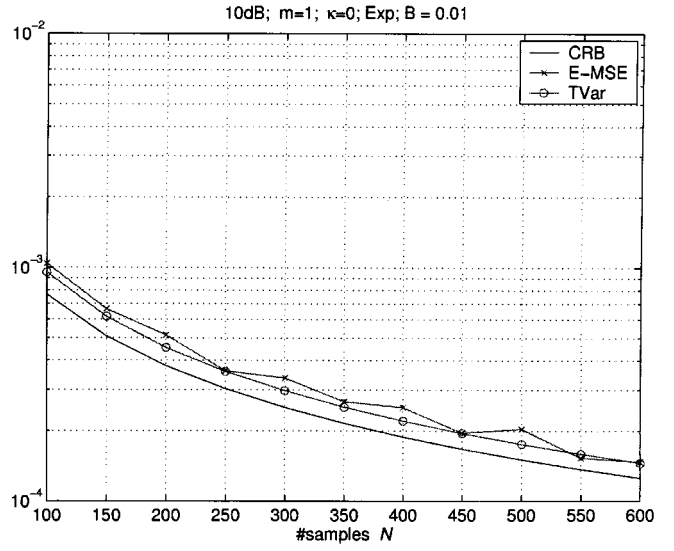


Fig. 6. Empirical MSE, theoretical variance, and CRB of $\omega^{(m)}$ versus N ; SNR = 10 dB; $\Delta_\omega = (2\pi)0.01$ (fast fading); $\kappa = 0$; $m = 1$.

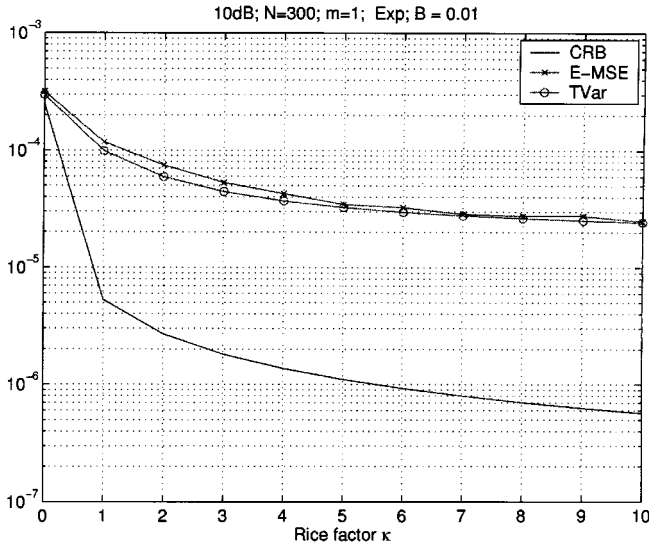


Fig. 5. Empirical MSE, theoretical variance, and CRB of $\omega^{(m)}$ versus κ ; SNR = 10 dB; $\Delta_\omega = (2\pi)0.01$ (fast fading); $m = 1$; $N = 300$.

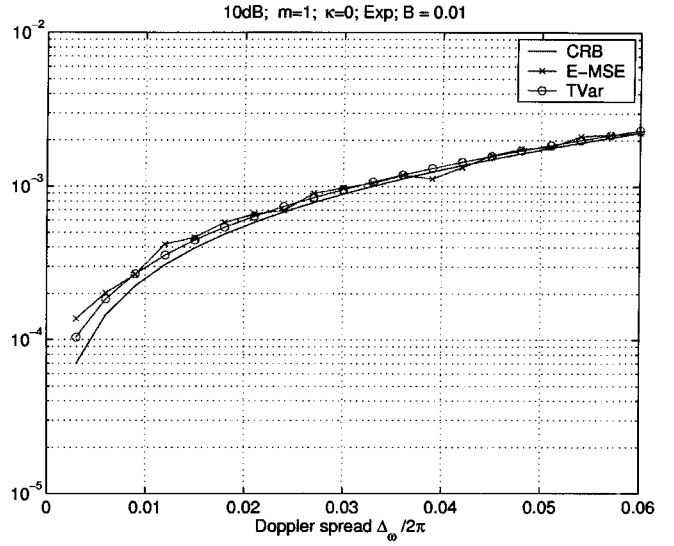


Fig. 7. Empirical MSE, theoretical variance, and CRB of $\omega^{(m)}$ versus Δ_ω ; SNR = 10 dB; $N = 300$; $\kappa = 0$; $m = 1$.

$N = 300$, $\kappa = 0$, SNR = 10 dB, $\omega = (2\pi)0.01$, and $m = 1$. The empirical and theoretical MSEs are in very good agreement in all cases. For $\kappa = 0$, the PP estimator is efficient as the SNR increases (see Fig. 4) or as Δ_ω increases (see Fig. 7) but not as κ increases (see Fig. 5). Indeed, Fig. 5 indicates that m should increase with κ ; indeed, if κ can be estimated, one could choose a better value of m . Fig. 9 shows the improved performance when $m = \kappa + 1$ was chosen; this choice is arbitrary and not optimal; optimal choice of m requires knowledge of $R_a(\tau)$, κ , and SNR. Finally, Fig. 8 demonstrates the empirical performance, and the theoretical bounds are independent of ω . The choice of $m = 1$ was motivated by the earlier examples; for $m = 10$, the behavior is similar, but the performance of the PP estimator is more offset from the CRB.

We report on other simulations for which we do not include figures due to lack of space. When $\kappa = 1$, the PP estimator with $m = 1$ is no longer statistically efficient as the SNR or Δ_ω

increases; indeed, as Δ_ω increases with $\kappa = 1$ fixed, the CRB decreases monotonically, whereas the empirical and theoretical variances increase monotonically! The theoretical and empirical variance expressions are in good agreement, but they are now farther away from the CRB. These results suggest that it would be useful to develop estimators of κ , Δ_ω , and SNR.

VI. CONCLUSIONS

We considered the estimation of the frequency of a complex sinusoid in the presence of circular complex-valued colored multiplicative and additive white noises. We derived closed-form expressions for the finite and large-sample CRBs. It was shown that in the presence of multiplicative noise, the CRB for the frequency may be much higher than that in the nonfading case, particularly in the Rayleigh case, where the CRB is $O(N^{-1})$ instead of $O(N^{-3})$, where N is the number of samples. We then derived expressions for the theoretical

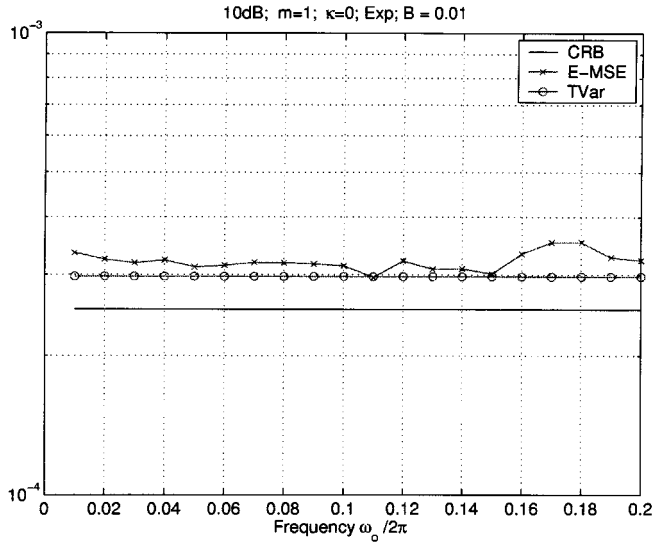


Fig. 8. Empirical MSE, theoretical variance, and CRB of $\omega^{(m)}$ versus ω_o ; SNR = 10 dB; $\Delta\omega = (2\pi)0.01$ (fast fading); $N = 300$; $\kappa = 0$; $m = 1$.

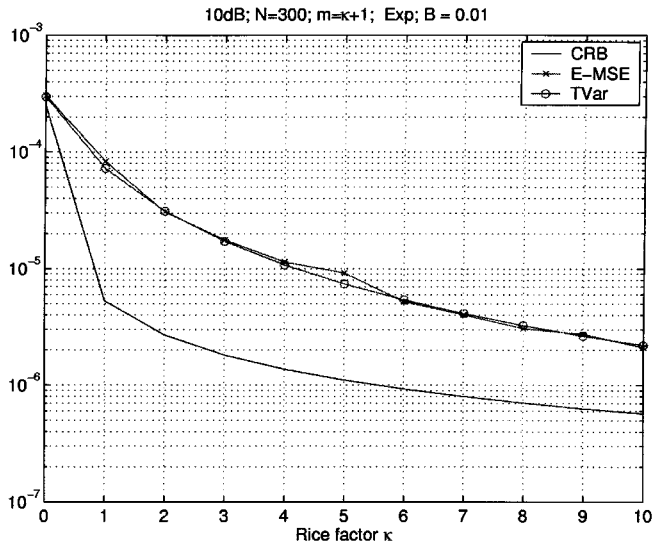


Fig. 9. Empirical MSE, theoretical variance, and CRB of $\omega^{(m)}$ versus κ ; SNR = 10 dB; $\Delta\omega = (2\pi)0.01$ (fast fading); $m = \kappa + 1$; $N = 300$.

variances of frequency estimators that are based on the sample correlations.

We derived finite- and large-sample expressions for the variance of the pulse-pair method, which estimates the frequency from the phase of the m th lag of the correlation. We saw that at high SNR, $m = 1$ is an appropriate choice, regardless of the values of the Rician factor (κ) and the Doppler spread ($\Delta\omega$). Indeed, when κ is large and/or $\Delta\omega$ is small, the performance of the PP estimator does not vary much with m . When κ is small and $\Delta\omega$ is large, the performance might degrade dramatically as m increases; in these cases, the performance of the $m = 1$ estimator is close to that of the optimal m . However, for low to moderate SNRs, the performance of the estimator varies significantly with the lag m ; when κ is large and/or $\Delta\omega$ is small, the optimal m is $m_o \approx 2N/3$. Otherwise, m_o lies between 1 and $2N/3$; the lower κ and the larger $\Delta\omega$ are, the smaller m_o will

be. Further, it was seen that with $m = m_o$, the performance of the PP estimator is close to the CRB.

APPENDIX A

COMPUTATION OF THE FINITE-SAMPLE FIM

For notational simplicity, we rewrite (3) as

$$J_{\theta_k, \theta_\ell} = J_m(\theta_m, \theta_\ell) + J_R(\theta_k, \theta_\ell)$$

where

$$J_m(\theta_k, \theta_\ell) = 2\mathcal{R} \left\{ \frac{\partial \mathbf{m}_x^H}{\partial \theta_k} \mathbf{R}_x^{-1} \frac{\partial \mathbf{m}_x}{\partial \theta_\ell} \right\}$$

$$J_R(\theta_k, \theta_\ell) = \text{tr} \left\{ \mathbf{R}_x^{-1} \frac{\partial \mathbf{R}_x}{\partial \theta_k} \mathbf{R}_x^{-1} \frac{\partial \mathbf{R}_x}{\partial \theta_\ell} \right\}.$$

The partial derivatives of $\mathbf{m}_x$ and $\mathbf{R}_x$ are given by

$$\frac{\partial \mathbf{m}_x}{\partial \omega_o} = j2\mu e^{j\phi} \mathbf{D} \Gamma \mathbf{1}, \quad \frac{\partial \mathbf{m}_x}{\partial \mu} = e^{j\phi} \Gamma \mathbf{1}$$

$$\frac{\partial \mathbf{m}_x}{\partial \phi} = j\mu e^{j\phi} \Gamma \mathbf{1}, \quad \frac{\partial \mathbf{R}_x}{\partial \omega_o} = j2 [\mathbf{D} \Gamma \mathbf{R}_a \Gamma^H - \Gamma \mathbf{R}_a \Gamma^H \mathbf{D}]$$

$$\frac{\partial \mathbf{R}_x}{\partial \sigma_v^2} = \mathbf{I}, \quad \frac{\partial \mathbf{R}_x}{\partial \alpha_k} = \Gamma \frac{\partial \mathbf{R}_a}{\partial \alpha_k} \Gamma^H, \quad k = 1, \dots, p \quad (21)$$

where $\mathbf{D} = \text{diag}\{0, 1, \dots, N-1\}$. Note that $\mathbf{R}_x$ is not a function of μ or ϕ , and similarly, $\mathbf{m}_x$ is not a function of α or σ_v^2 .

Computation of J_{ω_o, ω_o} : Using (21), $J_m(\omega_o, \omega_o)$ can be written as

$$J_m(\omega_o, \omega_o) = 2\mu^2 \mathbf{1}^T \mathbf{D} \Gamma^H \mathbf{R}_x^{-1} \Gamma \mathbf{D} \mathbf{1}$$

$$= 2\mu^2 \mathbf{1}^T \mathbf{D} (\mathbf{R}_a + \sigma_v^2 \mathbf{I})^{-1} \mathbf{D} \mathbf{1}$$

since $\Gamma^H \Gamma = \mathbf{I}$. Let $\mathbf{R} = \mathbf{R}_a + \sigma_v^2 \mathbf{I}$; using (21) yields

$$J_R(\omega_o, \omega_o) = \text{tr} \left\{ \Gamma \mathbf{R}^{-1} \Gamma^H (j \Gamma \mathbf{D} \mathbf{R}_a \Gamma^H - j \Gamma \mathbf{R}_a \mathbf{D} \Gamma^H) \right. \\ \left. \cdot \Gamma \mathbf{R}^{-1} \Gamma^H (j \Gamma \mathbf{D} \mathbf{R}_a \Gamma^H - j \Gamma \mathbf{R}_a \mathbf{D} \Gamma^H) \right\}.$$

Since $\mathbf{R}_a \mathbf{R}^{-1} = \mathbf{R}^{-1} \mathbf{R}_a$, and $\text{tr}(AB) = \text{tr}(BA)$, we obtain

$$J_R(\omega_o, \omega_o) = 2 \text{tr} \{ \mathbf{R}^{-1} \mathbf{D} \mathbf{R} \mathbf{D} \} - 2 \text{tr} \{ \mathbf{D}^2 \}.$$

Computation of $J_{\omega_o, \alpha}$: Since $\mathbf{m}_x$ is independent of α , $J_m(\omega_o, \alpha_k) = 0$, and

$$J_{\omega_o, \alpha_k} = J_R(\omega_o, \alpha_k), \quad k = 1, \dots, p.$$

Using (21), we have

$$J_{\omega_o, \alpha_k} = j \text{tr} \left\{ \mathbf{R}^{-1} \mathbf{D} \mathbf{R}_a \mathbf{R}^{-1} \frac{\partial \mathbf{R}_a}{\partial \alpha_k} \right\} \\ - j \text{tr} \left\{ \mathbf{R}^{-1} \mathbf{R}_a \mathbf{D} \mathbf{R}^{-1} \frac{\partial \mathbf{R}_a}{\partial \alpha_k} \right\}, \quad k = 1, \dots, p.$$

Since the trace operator is commutative, $\mathbf{R}_a \mathbf{R}^{-1} = \mathbf{R}^{-1} \mathbf{R}_a$, and $\mathbf{D}$ is real-valued, we obtain

$$J_{\omega_o, \alpha_k} = -2 \text{tr} \left\{ \mathbf{D} \frac{\partial \mathbf{R}_a}{\partial \alpha_k} \mathbf{R}^{-1} \right\}, \quad k = 1, \dots, p.$$

The derivation of the other FIM entries is straightforward and, hence, omitted.

APPENDIX B COMPUTATION OF THE LARGE-SAMPLE FIM

For large N , the first term of J_{ω_o, ω_o} is given by

$$J_m(\omega_o, \omega_o) = 2\mu^2 \mathbf{1}^T \mathbf{D} \mathbf{R}^{-1} \mathbf{D} \mathbf{1} \approx \frac{2\mu^2 N^3/3}{S_a(0) + \sigma_v^2} \quad (22)$$

where $S_a(\omega)$ is the PSD of $a(n)$. Since $x(n)$ is Gaussian and circular around its mean, the second term of the FIM expression in (3) is given by Whittle's formula for large N [9, p. 535]

$$2 \operatorname{tr} \left\{ \mathbf{R}_x^{-1} \frac{\partial \mathbf{R}_x}{\partial \theta_k} \mathbf{R}_x^{-1} \frac{\partial \mathbf{R}_x}{\partial \theta_\ell} \right\} \approx N \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{\frac{\partial S_x(\omega)}{\partial \theta_k} \frac{\partial S_x(\omega)}{\partial \theta_\ell}}{S_x^2(\omega)} d\omega \quad (23)$$

where $S_x(\omega) = S_a(\omega - \omega_o) + \sigma_v^2$ is the PSD of $x(n)$. Therefore, for large N , we have that

$$J_{\omega_o, \omega_o} \approx \frac{2\mu^2 N^3/3}{S_a(0) + \sigma_v^2} + \frac{N}{2\pi} \int_{-\pi}^{\pi} \left(\frac{S'_a(\omega)}{S_a(\omega) + \sigma_v^2} \right)^2 d\omega$$

where $S'_a(\omega) := (dS_a(\omega))/(d\omega)$.

The other FIM entries are similarly obtained.

APPENDIX C COVARIANCES OF SAMPLE ESTIMATES OF THE CORRELATION

Let $\operatorname{cum}(a) := \mathcal{E}\{a\}$, and $\operatorname{cum}(a, b) := \mathcal{E}\{ab\} - \mathcal{E}\{a\}\mathcal{E}\{b\}$; then, we have for jointly Gaussian random variables (a, b, c, d) , [8], [11]

$$\begin{aligned} \operatorname{cov}(ab, c^*d^*) &= \operatorname{cum}(ab, cd) = \mathcal{E}\{abcd\} - \mathcal{E}\{ab\}\mathcal{E}\{cd\} \\ &= \operatorname{cum}(a, c)\operatorname{cum}(b, d) + \operatorname{cum}(a, d)\operatorname{cum}(b, c) \\ &\quad + \operatorname{cum}(a)\operatorname{cum}(c)\operatorname{cum}(b, d) \\ &\quad + \operatorname{cum}(a)\operatorname{cum}(d)\operatorname{cum}(b, c) \\ &\quad + \operatorname{cum}(b)\operatorname{cum}(d)\operatorname{cum}(a, c) \\ &\quad + \operatorname{cum}(b)\operatorname{cum}(c)\operatorname{cum}(a, d). \end{aligned} \quad (24)$$

We will derive the covariance of the correlation estimate in (7). Assume, without loss of generality, that $\ell \geq m \geq 0$. We can express the sample estimate as

$$\begin{aligned} \hat{c}_x(m) &= \frac{1}{N-m} \sum_{n=0}^{N-1-m} x^*(n)x(n+m) \\ &= \frac{e^{j\omega_o m}}{N-m} \sum_{n=0}^{N-1-m} y^*(n)y(n+m) \end{aligned}$$

where

$$y(n) := x(n) \exp(-j\omega_o n) = a(n) + v(n) \exp(-j\omega_o n)$$

is a stationary Gaussian process with mean $\mu \exp(j\phi)$ and covariance $R_y(m) = R_a(m) + R_v(m) = R_a(m) + \sigma_v^2 \delta_m$. Hence

$$\begin{aligned} \operatorname{cov}\{\hat{c}_x(m), \hat{c}_x(\ell)\} &= \frac{e^{j\omega_o(m-\ell)}}{(N-m)(N-\ell)} \sum_{n=0}^{N-1-m} \sum_{t=0}^{N-1-\ell} \\ &\quad \cdot \operatorname{cov}\{y^*(n)y(n+m), y^*(t)y(t+\ell)\}. \end{aligned}$$

With $a = y^*(n)$, $b = y(n+m)$, $c = y(t)$, and $d = y^*(t+\ell)$, we see that the second, fourth, and sixth terms of (24) are zero because of the circular symmetry of $y(n)$. Hence

$$\begin{aligned} &e^{-j\omega_o(m-\ell)}(N-m)(N-\ell) \operatorname{cov}\{\hat{c}_x(m), \hat{c}_x(\ell)\} \\ &= \sum_{n=0}^{N-1-m} \sum_{t=0}^{N-1-\ell} R_y(t-n)R_y(n-t+m-\ell) \\ &\quad + \mu^2 R_y(t-n) + \mu^2 R_y(n-t+m-\ell) \\ &= \sum_{\tau=m-N+1}^{N-1-\ell} D(\tau) [R_y(\tau)R_y^*(\tau+\ell-m) + \mu^2 R_y(\tau) \\ &\quad + \mu^2 R_y^*(\tau+\ell-m)] \end{aligned} \quad (25)$$

where, for $-(N-1-m) \leq \tau \leq N-1-\ell$

$$D(\tau) = \max(N-\ell-|\tau| - \min(0, \max(\tau, m-\ell)), 0). \quad (26)$$

In a similar fashion, we have

$$\begin{aligned} \operatorname{cov}\{\hat{c}_x(m), \hat{c}_x^*(\ell)\} &= \frac{e^{j\omega_o(m+\ell)}}{(N-m)(N-\ell)} \sum_{n=0}^{N-1-m} \sum_{t=0}^{N-1-\ell} \\ &\quad \cdot \operatorname{cov}\{y^*(n)y(n+m), y(t)y^*(t+\ell)\}. \end{aligned}$$

With $a = y^*(n)$, $b = y(n+m)$, $c = y^*(t)$, and $d = y(t+\ell)$, we see that the first, third, and fifth terms of (24) are zero because of the circular symmetry of $y(n)$. Hence

$$\begin{aligned} &e^{-j\omega_o(m+\ell)}(N-m)(N-\ell) \operatorname{cov}\{\hat{c}_x(m), \hat{c}_x^*(\ell)\} \\ &= \sum_{\tau=m-N+1}^{N-1-\ell} D(\tau) [R_y(\tau+\ell)R_y^*(\tau-m) + \mu^2 R_y(\tau+\ell) \\ &\quad + \mu^2 R_y^*(\tau-m)]. \end{aligned} \quad (27)$$

For our model, we have $R_y(\tau) = R_a(\tau) + \sigma_v^2 \delta(\tau)$ with $R_a(\tau)$ real valued. From (26), we have

$$D(0) = D(m-\ell) = N-\ell,$$

$$D(m) = D(-\ell) = \max(N-\ell-m, 0).$$

Using these in (25) and (27) leads to Proposition 3.

APPENDIX D COVARIANCE OF PHASE ESTIMATES

Let $c_x(m) = |c_x(m)| \exp(j\varphi(m))$, and consider its logarithm

$$\ln c_x(m) = \ln |c_x(m)| + j\varphi(m).$$

Differentiating both sides of this equation gives

$$\delta\varphi(m) = -j \frac{\delta\{c_x(m)\}}{c_x(m)} + j \frac{\delta\{|c_x(m)|\}}{|c_x(m)|}.$$

Since $\delta\varphi(m)$ is real-valued, we obtain

$$\delta\varphi(m) = \mathcal{I} \left\{ \frac{\delta\{c_x(m)\}}{c_x(m)} \right\}$$

which yields

$$\begin{aligned} \text{cov}\{\hat{\varphi}(m), \hat{\varphi}(\ell)\} &= \frac{1}{2} \mathcal{R} \left\{ \frac{\text{cov}\{\hat{c}_x(m), \hat{c}_x(\ell)\}}{c_x(m)c_x^*(\ell)} - \frac{\text{cov}\{\hat{c}_x(m), \hat{c}_x^*(\ell)\}}{c_x(m)c_x(\ell)} \right\}. \end{aligned} \quad (28)$$

Substituting from (8) and (9) into (28), we obtain [noting that $R_y(\tau)$ is real-valued]

$$\begin{aligned} \text{cov}\{\hat{\varphi}(m), \hat{\varphi}(\ell)\} &= \frac{1}{2(N-m)(N-\ell)} \sum_{\tau=m-N+1}^{N-1-\ell} D(\tau) \\ &\cdot \frac{D(\tau)}{(R_y(m) + \mu^2)(R_y(\ell) + \mu^2)} \\ &\cdot [R_y(\tau)R_y(\tau + \ell - m) + \mu^2 R_y(\tau) \\ &\quad + \mu^2 R_y(\tau + \ell - m) - R_y(\tau + \ell)R_y(\tau - m) \\ &\quad - \mu^2 R_y(\tau + \ell) - \mu^2 R_y(\tau - m)]. \end{aligned}$$

For our model, $R_y(\tau) = R_a(\tau)$, $\tau \neq 0$; in addition, $R_a(\tau)$ is real-valued. We use this, along with the previously defined Rice factor, SNR, and normalized correlation, to obtain Proposition 5.

APPENDIX E IMPROVED VARIANCE EXPRESSIONS

In deriving the variance expression for the phase estimator, we used the first-order Taylor expansion of the log function

$$\ln \frac{\hat{c}_x(m)}{c_x(m)} = \ln \frac{1 - \tilde{c}_x(m)}{c_x(m)} \approx \frac{\tilde{c}_x(m)}{c_x(m)}$$

which is valid only if $|\tilde{c}_x(m)| \ll |c_x(m)|$; in the case of strong temporal correlation, the errors may be large for finite N . Therefore, one might want to use the next term in the Taylor expansion

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} \dots$$

assuming $-1 < x \leq 1$.

Let

$$\frac{\hat{c}_x(m)}{c_x(m)} = \frac{c_x(m) - \tilde{c}_x(m)}{c_x(m)} = 1 - y.$$

We have

$$\begin{aligned} \hat{\varphi}(m) &= \mathcal{I} \ln \frac{\hat{c}_x(m)}{c_x(m)} \\ &= \mathcal{I} \ln(1-y) \approx \mathcal{I} \left(y - \frac{1}{2} y^2 - \frac{1}{3} y^3 \dots \right). \end{aligned}$$

Hence

$$\text{var}_2(\hat{\varphi}(m)) \approx \mathcal{E}\{|y|^2\} + \frac{1}{4} \mathcal{E}\{|y|^4\} - 2\mathcal{R}\mathcal{E}\{y|y|^2\}.$$

For large N , y is Gaussian distributed. For small N , this assumption may not hold. Invoking this assumption, and noting that y is complex, we have

$$\text{var}_2(\hat{\varphi}(m)) \approx \mathcal{E}\{|y|^2\} \left(1 + \frac{1}{2} \mathcal{E}\{|y|^2\} \right)$$

where $\mathcal{E}|y|^2$ is given by $\text{cov}(\hat{\varphi}(m), \hat{\varphi}(m))$ in Propositions 5 and 6. This var_2 expression was used to plot the theoretical MSEs in the figures.

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